

Rahul Shamkuwar

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EXPERIENCE

• Workday

Sr. Associate Software Development Engineer

Boulder, CO

Feb 2025 - Present

- **AI Agent Observability & Evaluation Platform (Python)**: Architected and productionized Aval, a Python framework for evaluating LLM-powered AI agents via async, graph-based evaluations integrated with LangSmith. Engineered CI/CD packaging pipelines to accelerate developer iteration.
- **Finance Agent Development (FastAPI/LangGraph)**: Scaffolded fin-procurement-agent using the ReAct pattern in agent-forge; built and scaled FastAPI services powering Self-Service, AP Intelligence, and Procurement finance AI agents, integrating the LangGraph runtime, external APIs, and RDS PostgreSQL.
- **Agent Architecture & Graph Design**: Designed and validated a POC for a custom stateless LangGraph backend, decoupling the Procurement agent into an independent graph; implemented human-in-the-loop (HITL) intervention capabilities for agent workflows.
- **Full-Stack Product Development (React/TS/Python/PostgreSQL)**: Led end-to-end delivery of the GenAI Studio observability & evals interface (React/TypeScript + Python), designing PostgreSQL models that power complex evaluation metrics at scale.
- **Cross-Team Technical Leadership (Hive Integration)**: Led cross-team technical discovery with the Hive/Resolution Agent team on permissioning, versioning, and intent classification; authored proposals that resolved integration blockers and aligned engineering teams on a shared path forward.

• BioFrontiers Institute

Software Engineer

Boulder, CO

May 2024 - Jan 2025

- **Full-Stack System Modernization (React/Next.js/TypeScript)**: Rebuilt a legacy web application with Next.js, TypeScript, MUI, and PostgreSQL, cutting maintenance complexity by 30%.
- **API Design & Performance Optimization**: Designed and optimized backend API routes and SSR components for data-intensive workloads, improving large-dataset delivery by 25%.
- **Data Storage & Automated Deployments**: Migrated large analytical datasets in PostgreSQL with 100% integrity and automated Docker Compose deployments, cutting deploy time by 50%.

• Workday

Software Development Engineer Intern

Boulder, CO

May 2024 - Aug 2024

- **Automated Releases & CI/CD**: Parallelized release pipelines and test automation in GitHub Actions, reducing pipeline execution time by 15% and setup times by 25%.
- **External Service Integration**: Built service interfaces to 6 external LLM providers, standardizing diverse responses and improving system stability by 13%.

• Trimble Cloud

Software Engineer

Boulder, CO

Sep 2023 - May 2024

- **Large-Scale Distributed Systems**: Established a Kubernetes cluster for microservices and deployed AWS Lambda functions, increasing processing efficiency by 20% and reducing latency.
- **Cloud Infrastructure & Automation**: Architected distributed AWS applications with Terraform (IaC), accelerating deployment provisioning by 30%.

SKILLS

- **Languages**: Python, TypeScript, JavaScript, SQL (PostgreSQL), Java, C++, HCL (Terraform), HTML/CSS.
- **Technologies**: LangGraph, React, Next.js, Node.js, FastAPI, Flask, REST APIs, Docker, Kubernetes, AWS, CI/CD (GitHub Actions).
- **Data & AI**: PostgreSQL, RDS, Prisma, pandas, NumPy, LLMs, LangGraph, Multi-Agent Systems, AI Agent Evaluation, Observability.
- **Core Competencies**: Full-Stack Development, Agent Architecture Design, API Design & Scaling, Distributed Systems, Cross-Team Technical Leadership, System Design.

EDUCATION

• University of Colorado Boulder

Bachelor of Science in Computer Science; GPA: 3.96

Boulder, CO

• University of Illinois Urbana-Champaign

Bachelor of Science in Aerospace Engineering; GPA: 3.46

Urbana-Champaign, IL